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System Properties Comparison Cassandra vs. CouchDB vs. MongoDB

Please select another system to include it in the comparison.

Editorial information provided by DB-Engines

Name	Cassandra X	CouchDB X	MongoDB X
Description	Wide-column store based on ideas of BigTable and DynamoDB	A native JSON - document store inspired by Lotus Notes, scalable from globally distributed server-clusters down to mobile phones.	One of the most popular document stores available both as a fully managed cloud service and for deployment on self-managed infrastructure
Primary database model	Wide column store	Document store	Document store
Secondary database models		Spatial DBMS	Spatial DBMS Search engine Time Series DBMS
DB-Engines Ranking Trend Chart	Score 115.45 Rank #11 Overall #1 Wide column stores	Score 17.16 Rank #41 Overall #7 Document stores	Score 480.73 Rank #5 Overall #1 Document stores
Website	cassandra.apache.org	couchdb.apache.org	www.mongodb.com
Technical documentation	cassandra.apache.org/doc/latest	docs.couchdb.org/en/stable	docs.mongodb.com/manual
Developer	Apache Software Foundation	Apache Software Foundation	MongoDB, Inc
Initial release	2008	2005	2009
Current release	4.0.4, May 2022	3.2.2, April 2022	5.0.8, April 2022
License	Open Source	Open Source	Open Source
Cloud-based only	no	no	no
DBaaS offerings (sponsored links)	Astra DB: Multi-cloud DBaaS built on Apache Cassandra.		• MongoDB Atlas: Global multi-cloud database with unmatched data distribution and mobility across AWS, Azure, and Google Cloud, built-in automation for resource and workload optimization, and so much more. • ScaleGrid for MongoDB: Fully managed hosting for MongoDB database on AWS, Azure and DigitalOcean with high availability and SSH access on the #1 multi-cloud DBaaS.
Implementation language	Java	Erlang	C++

Server operating systems	BSD Linux OS X Windows	Android BSD Linux OS X Solaris Windows	Linux OS X Solaris Windows
Data scheme	schema-free	schema-free	schema-free
Typing	yes	no	yes
XML support	no	no	
Secondary indexes	restricted	yes	yes
SQL	SQL-like SELECT, DML and DDL statements (CQL)	no	Read-only SQL queries via the MongoDB Connector for BI
APIs and other access methods	Proprietary protocol Thrift	RESTful HTTP/JSON API	proprietary protocol using JSON
Supported programming languages	C# C++ Clojure Erlang Go Haskell Java JavaScript Perl PHP Python Ruby Scala	C C# ColdFusion Erlang Haskell Java JavaScript Lisp Lua Objective-C OCaml Perl PHP PL/SQL Python Ruby Smalltalk	Actionscript C C# C++ Clojure ColdFusion D Dart Delphi Erlang Go Groovy Haskell Java JavaScript Lisp Lua MatLab Perl PHP PowerShell Prolog Python R Ruby Rust Scala Smalltalk Swift
Server-side scripts	no	View functions in JavaScript	JavaScript
Triggers	yes	yes	yes
Partitioning methods	Sharding	Sharding	Sharding
Replication methods	selectable replication factor	Multi-source replication Source-replica replication	Multi-Source deployments with MongoDB Atlas Global Clusters Source-replica replication
MapReduce	yes	yes	yes
Consistency concepts	Eventual Consistency Immediate Consistency	Eventual Consistency	Eventual Consistency Immediate Consistency
Foreign keys	no	no	no
Transaction concepts	no	no	Multi-document ACID Transactions with snapshot isolation
Concurrency	yes	yes	yes

Durability	yes	yes	yes
In-memory capabilities	no	no	yes
User concepts	Access rights for users can be defined per object	Access rights for users can be defined per database	Access rights for users and roles

More information provided by the system vendor

	Cassandra	CouchDB	MongoDB
Specific characteristics	Apache Cassandra is the leading NoSQL, distributed database management system, well... » more		MongoDB's application data platform provides developers a unified interface to power... » more
Competitive advantages	No single point of failure ensures 100% availability . Operational simplicity for... » more		The MongoDB platform can be used by developers building transactional, operational,... » more
Typical application scenarios	Internet of Things (IOT), fraud detection applications, recommendation engines, product... » more		Internet of Things and Time Series (Bosch, Jaguar Land Rover) Mobile (The Weather... » more
Key customers	Apple, Netflix, Uber, ING,, Intuit,Fidelity, NY Times, Outbrain, BazaarVoice, Best... » more		ADP, Adobe, Amadeus, AstraZeneca, Auto Trader, Barclays, BBVA, Bosch, Cisco, CERN,... » more
Market metrics	Cassandra is used by 40% of the Fortune 100. » more		200+ million downloads, over 1.5M Atlas database clusters under management handling... » more
Licensing and pricing models	Apache licenseÂ Pricing for commercial distributions provided by DataStax and available... » more		MongoDB database server: Server-Side Public License (SSPL) . Commercial licenses... » more
News	How to Web Scrape on a Schedule with Apache Cassandra, FastAPI, and Python 2 June 2022 Advanced Data Modeling on Apache Cassandra 31 May 2022 DataStax Employee Spotlight: Deepak Anand 30 May 2022 Create Cassandra Apps on AWS in Minutes, Not Weeks 25 May 2022 Cassandra Data Loading: 8 Tips for Loading Data into Astra DB 24 May 2022		

We invite representatives of system vendors to contact us for updating and extending the system information, and for displaying vendor-provided information such as key customers, competitive advantages and market metrics.

Related products and services

3rd parties	Instaclustr: Hosted & Managed Apache Cassandra as a Service » more CData: Connect to Big Data & NoSQL through standard Drivers. » more DataStax Cassandra: Zero touch operations, configuration and management with true autoscaling » more		CData: Connect to Big Data & NoSQL through standard Drivers. » more Navicat for MongoDB gives you a highly effective GUI interface for MongoDB database management, administration and development. » more Percona: Database problems? Not on your watch. Databases run better with Percona. » more

We invite representatives of vendors of related products to contact us for presenting information about their offerings here.

More resources

	Cassandra	CouchDB	MongoDB
DB-Engines blog posts	Cassandra keeps climbing the ranks of the DB-Engines Ranking 3 May 2016, Matthias Gelbmann Oracle is the DBMS of the Year 5 January 2016, Paul Andlinger, Matthias Gelbmann Winners, losers and an attractive newcomer in Novembers DB-Engines ranking 2 November 2015, Paul Andlinger show all	Couchbase climbs up the DB-Engines Ranking, increasing its popularity by 10% every month 2 June 2014, Matthias Gelbmann show all	Snowflake is the DBMS of the Year 2021 3 January 2022, Paul Andlinger, Matthias Gelbmann PostgreSQL is the DBMS of the Year 2020 4 January 2021, Paul Andlinger, Matthias Gelbmann PostgreSQL is the DBMS of the Year 2018 2 January 2019, Paul Andlinger, Matthias Gelbmann show all
Recent citations in the news	GraphQL and NoSQL Databases: The Type Mismatch 27 May 2022, thenewstack.io Couchbase Server 7.1 advances enterprise NoSQL database 10 May 2022, TechTarget NoSQL, NoMQ: Palo Alto Networks' New Event Streaming Paradigm 24 May 2022, thenewstack.io DataStax Reports Strong Q1 Momentum with Global Adoption of its Astra DB Cloud Database 31 May 2022, Valdosta Daily Times This is the beginning of the unbundled database era 1 June 2022, TechCrunch provided by Google News	IBM Cloudant pulls plan to fund new foundational layer for CouchDB 15 March 2022, The Register NoSQL Software Market Size, Scope and Forecast | Amazon, Couchbase, MongoDB Inc., Microsoft, Marklogic, OrientDB, ArangoDB, Redis, CouchDB, DataStax. – The Greater Binghamton Business Journal 3 June 2022, The Greater Binghamton Business Journal CouchDB 3.2 Improves Replication Scheduler 21 October 2021, iProgrammer Syncing Hyperledger Fabric StateDB (CouchDB) to Elasticsearch - Latest Digital Transformation Trends | Cloud News 4 November 2021, Wire19	Stocks making the biggest moves midday: Microsoft, Chewy, GameStop, MongoDB & more 2 June 2022, CNBC MongoDB just opened a hybrid Philly office for its local employees 11 May 2022, Technical.ly Why Shares of MongoDB Tanked 33% in May 3 June 2022, The Motley Fool Why MongoDB Popped 14.7% This Week 3 June 2022, Nasdaq What to expect during the MongoDB World event: Join theCUBE June 7 1 June 2022, SiliconANGLE News provided by Google News

		How to Connect Your Flask App With CouchDB: A NoSQL Database - MUO 14 August 2021, MakeUseOf provided by Google News	
Job opportunities	Cassandra Database administrator AT&T, Hyderabad Cassandra Database administrator AT&T, Chennai MCS- Database Technologies - Senior Associate 1 PwC, Bengaluru Senior Hadoop/Cassandra Administrator Infosys Limited, Bengaluru Database Administrator Fidelity Investments, Chennai	Web Developer Windy Cloud Technologies, Coimbatore Specialist Programmer-MEAN/MERN Infosys Limited, Bengaluru Full Stack Developer Wits Innovation Lab, Kharar SQL Fresher VASPP TECHNOLOGIES, Bengaluru Python Developer DCUBE Ai, Thiruvananthapuram	Web Developer Ontoborn Technologies LLC, Chennai Web Developer Windy Cloud Technologies, Coimbatore Junior PHP Developer Verve Mobile Private Limited, Hyderabad Full Stack Developer TrueLedge (HR and Accounting Services), Coimbatore Backend Developer DIGICOLLECT INDIA PRIVATE LIMITED, Bengaluru jobs by indeed

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